

Nightingale — technical brief

main · 3 September 2026 · synthetic data only

Measured on this tree: backend pytest **521 passed / 1 skipped**; frontend vitest **134 passed**; ruff, ruff-format, mypy and ty clean. **11 of 16 clinic scenarios survive, 5 are partial, none fail outright**. Seven are demonstrated on video, each gated by an automated check that asserts the on-screen evidence before any footage is kept. Per-scenario coverage is indexed in `SCENARIO_TEST_MAP.md`, which a test verifies against the source tree. Architecture detail is in `ARCHITECTURE.md`.

Where a claim rests on something I could not measure, it is listed as not evidenced rather than described as working.

1 · What the feedback changed

The feedback's central move was to stop asking whether a feature exists and start asking what a clinician sees when it misbehaves. Re-reading the build through that lens produced four patterns, none of which I expected.

The safety machinery was consistently ahead of the surfaces consuming it. The evaluation harness had really run and its negative result really did gate the system; anchors were frozen by trigger. What was missing, repeatedly, was the last hop: a reason-label map covering two of six backend values, a `fallback_kind` badge that was unreachable dead code, a `current_confidence_state` computed on the server and then re-derived — wrongly — in the browser, a diff endpoint with no caller. A reviewer reading only the backend would have scored this build higher than a clinician using it.

Absence was being read as innocence. Four trust defects reduced to one mistake: a missing `DecisionAssessment` row was treated as "nothing to qualify" rather than "not qualified". That reading let a job report qualified confidence from a partial subset, let an unassessed model-derived highlight render as a ready priority and reach the patient projection, and let publication record a model claim as human-asserted. The fix is one explicit rule — a highlight is model-derived if it carries a candidate fingerprint, and a model-derived highlight with no assessment is unqualified — threaded through job confidence, glance review state and the publication gate by two shared helpers instead of three divergent checks.

The same mistake reappeared in an unrelated subsystem. The row-level security write check on provenance pointers validated a row by querying for that row, which cannot be satisfied before the row exists. A not-yet-present thing treated as a fact about the world rather than as a gap in what is known.

State machines right in the model and wrong at the boundary. A delivered SMS could be marked revoked, which is a false record — except for enrolment and one-time codes, where revocation is precisely how the live token is invalidated. Clinic preflight returned a 500 on exactly the out-of-range retention configuration it exists to report.

2 · Scored build parts

Survives — seven of twelve. *Medical terminology and dosage confirmation*: a versioned RxNorm-coded clinic formulary screened at publication, with four-axis human attestation enforced on client and server; the vocabulary is deliberately small and fails closed outside it. *Immutable, version-bound provenance*: triggers freeze anchors and seal version ciphertext, and append-only tables `REVOKE` `UPDATE`, `DELETE` from the runtime role. *Fact extraction under negation, correction and conflict*: three-valued polarity, four negation-scope guards, and unknown never becomes reassuring. *Real-time collaborative editing*: lost-update prevention proven under thread contention, with comment mutations carrying revisions and requiring `If-Match`. *AI regeneration preserving human state*: candidate fingerprints make regeneration reuse rather than duplicate, and confirmed state is protected by trigger in scoring, decay and cold storage. *Contradictory human, patient and AI assertions*: no source-precedence auto-resolution, and the conflict card names each side's origin, role, section and language. *Distinct clinician, staff and patient outputs*: server-side read and write gating, different response schemas per audience, a separate patient route tree.

Partial — five of twelve. *Streaming and noisy-environment ASR* uses a real SNR estimate rather than the earlier RMS proxy, but there is no denoiser and no measurement on real noisy recordings. *Diarization* runs pyannote in a killable subprocess with typed unavailability codes; no weights are cached here, so quality is unmeasured. *Within-statement code-switching* determines language per span at a 0.85 threshold and fails closed below it, but only against synthetic spans. *Multilingual downstream processing* gained Malay and Chinese dose patterns plus a language-agnostic numeric fallback, closing an English-only publication-gate bypass; there is still no translation step and no real-speech evaluation. *Self-learning* is clinic-scoped and bounded — ± 0.20 clamp by database check, $1/\sqrt{n}$ damping, append-only audit enforced by `REVOKE`, shadow mode by default, allergy and medication floors in both code and schema — but exposure is evaluated and **not corrected**: no exploration, no propensity weighting.

One deliberate omission inside a survivor. Nothing here generates for reading level. The patient summary is written by the publishing clinician. Automatic rewriting would put unreviewed model wording into the patient-visible record, contradicting the rule that human confirmation gates everything outbound. The audience separation is real; audience-adapted generation is not claimed.

3 · Where this fails first

Message delivery. No Twilio credentials exist here, so the effective provider is a deterministic stub that reports success. Scenarios 1, 5 and 11 all terminate at the same unproven hop. One sandbox account moves three scenarios at once; without it, a queued message looks delivered, which is the failure mode that matters.

The voice path under a hang. Text jobs are bounded at 15/30/75 seconds and fall back deterministically. Voice has only a 600-second per-call timeout, so a stuck transcript shows an indefinite spinner with no elapsed time and no retry. This is the least-improved scenario and the one most likely to be hit in a real clinic.

Real multilingual speech. Every code-switching result here is a deterministic fixture. Hokkien is the weakest leg: the `nan` lexicon assumes POJ romanisation and Whisper does not emit POJ, so that branch is effectively unreachable from a real recording. There is no public Malay/English/Hokkien medical consult corpus to evaluate against. What the seven-agent consult pipeline was measured against instead, and where it still fails, is set out in `docs/TRILINGUAL_MULTI_AGENT_BRIEF.md`.

Development-mode escape hatches. The restricted-runtime-role assertion is skipped when `FASTAPI_ENV=development`. A staging box left in development mode with owner credentials loses the tenant-isolation guarantee that the rest of the design depends on.

4 · What I tried that did not work

Fixing tests before understanding them. The full backend suite gave 140 failures. Columns had been added to already-applied migrations, so the database was stamped at head while lacking them. Rebuilding the schema from empty reduced 140 failures to 8 — 132 environmental, 8 real signal. The rule that came out of it: no further edits to an applied migration.

Adding `entity:medication` to the protected-visibility predicate. It belongs in the learning safety floor, and it is now there with a matching database check constraint. Putting it in the visibility predicate as well would have reclassified every medication highlight into the clinical-review surface and silently changed dismissal permissions. Learning suppression and review classification are different concerns that happened to share a predicate.

A demo proof string that proved nothing. My first scenario-13 assertion was `"Penicillin Allergy"`, which passed while the conflict card did not exist at all: Playwright's substring match is case-insensitive and had matched the note body I had just typed. Proof strings are now unique to the card under test.

Dropped from this freeze: `VOICE_JOB_TIMEOUT_SECONDS` with elapsed-time and retry UI; the `LocalNormalizedText` nominal type; the original-versus-current diff pane; a persisted `MedicationReviewCandidate` table; exposure-bias correction. Each is named rather than quietly omitted.

5 · Assumptions

Still standing. Provenance must be version-bound and immutable; this held under every probe. The database, not the route layer, is where tenant isolation belongs — deleting a route's clinic filter still yields zero cross-boundary rows. Redaction ordering must be structural, so tampered text fails by construction rather than by convention. Shadow-mode-by-default for learning, which the exposure-bias critique confirmed rather than undermined. Human confirmation gates patient-facing dosage.

No longer standing.

That a green suite means a user-reachable path works. The suite was green while the patient portal returned a 500 on its primary action, because no test exercised a patient writing clinically meaningful text. The bug was found by trying to film the feature. Coverage measured the paths I thought to write, and my blind spot in the tests matched my blind spot in the code.

That a stubbed provider is an acceptable default. It reports success, so a queued message looks delivered. The state machine was never the problem; the honest default was.

That a declared config value is an enforced one. Retention days, messaging channels and a worker-enabled flag were stored, plumbed and unread. A setting that exists only to satisfy a test reads as a control during review and is worse than no setting.

That backend correctness implies clinical correctness. Correct data reaching the wire and never rendered is not a feature. The dead badge and the unreachable diff pane are the same failure as a missing check.

That "no assessment" is neutral. The assumption behind all four trust defects and behind the provenance write check.

That the demo environment reflects the code. It did not, silently, for eight hours, because an old image and an old schema agreed with each other.

6 · The demo is a test that produces footage

Each scenario is defined once with its click path and the strings that must be visible for it to be true. A check phase walks that path with capture off and asserts every string; the record phase replays it with capture on. A scenario that cannot prove itself produces no video, so footage cannot drift from the claim.

Filming scenario 13 found three defects, none reachable from the suite as written. Conflict cards showed UUID-sort ordinals — "first" versus "conflicting" assertion — instead of nurse versus patient, because origin was checked before section. A patient insight mentioning an allergy writes a provenance pointer, and that table's write check validated a pointer by querying for that pointer: a self-referential EXISTS, unsatisfiable before the row is visible, so **any patient submitting clinically meaningful text through their own portal got a 500** — invisible to clinical roles, whose branch short-circuits first. With that fixed, the patient's "no known drug allergies" was refused as an unresolved conflict, because the gate stopping clinician content going out to a patient was also stopping the patient's own input. All three now have regression tests.

Evaluation evidence, negative results kept. Fact extraction calibrates to low at a 0.0437 accuracy lower bound over 176 judged facts; voice to low with a 0.200 word error rate over 2,206 segment decisions; redaction achieves 1.0 PHI recall over 500 samples and passes. The first two are why the system abstains on every AI fact extraction today. Left exactly as measured.